



A SIMPLE RESERVOIR SIMULATION

CHEN 629 – Transport Phenomena



HUSNAIN MANZOOR

NOUR BASHA

GROUP 2

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Abstract

The aim of this report is to simulate a simple reservoir where a series of pressure measurements are given out at the three boundaries (the North, South, East and West side being impermeable), and calculate the two-dimensional pressure distribution of the oil inside that reservoir, in addition to presenting its respective contour plot, and the volumetric flux and fluid velocity vector plots. In this report, these calculations are carried out using Darcy's equation of fluid flow through a porous medium, derived for two-dimensional flow in the x - and the y -direction. In addition, the finite difference method is used to solve the resulting bivariate differential equation.

1 Introduction

To describe the fluid flow within porous medium, Henry Darcy conducted experiments of water flow through unconsolidated sand beds, taken as the porous medium, in a vertical tank. Pressurized water was then added to the top of the tank and allowed to flow out of the bottom. After some time, all the pores were filled with water and the inflow rate matched the outflow rate. Moreover, the water pressure at each end of the tube was expressed in terms of the respective height of the water in the tank. Consequently, this resulted in Darcy's Law. [1]

Darcy's Law was later expanded to include several types of fluids and movement within other media as well. As a result, Darcy could deduce that volumetric flow rate out of the tank (Q) is directly proportional to the cross-sectional area (A) perpendicular to the fluid's flow and to the difference in the height of the water ($H_2 - H_1$), where $H_2 < H_1$, and is inversely proportional to the flow length through the porous material (L). Similar conclusions can be derived for the flow of fluid in horizontal, vertical or inclined mediums. Darcy's Law is given by Equation (1) below, where P_b and P_a are the fluid pressures at the outlet and inlet of the porous medium respectively.

$$Q = \frac{-\kappa A (P_b - P_a)}{\mu L} \quad (1)$$

Furthermore, κ is the permeability and μ is the viscosity of the fluid.

Reservoir engineering and simulation considers several factors that affect the fluid flow in a reservoir and predicts the overall pressure gradients, volumetric and velocity fluxes within the reservoir. In the following report, a simple bivariate differential equation was first derived based on Darcy's Law, and it was then extended to describe the flow of fluid in our problem statement. Generally, the solutions to the derived differential equations would require a lot of time if solved

analytically; hence, a more sophisticated method of simulation was used and a MATLAB code was written to calculate the pressures at various nodes/points within the well. This technique is widely used to simulate fluid pressures inside a reservoir, since it requires minimal execution time and offers reasonably accurate results depending on the resolution of the pressure grid within the reservoir.

2 Assumptions for applying Darcy's Equation

The following assumptions were made for deriving the Darcy's equation:

1. The reservoir is a square prism with a square base in the xy -plane.
2. The flow is continuous and steady.
3. No chemical reactions within the reservoir.
4. The cross-sectional area perpendicular to the flow of the fluid (A) is parallel to the yz -plane.
5. The porosity (ε), which is the "ratio of pore volume to total volume", is 35%.
6. The permeability (κ) of is isotropic, meaning it is uniform in all directions and has a value of $3.5 \cdot 10^{-10} \text{ m}^2$.
7. Pressure changes in the z -direction (the depth of the reservoir) are ignored since the objective is to analyze the problem in two dimensions.
8. Steady state flow within the reservoir means no accumulation of oil within the reservoir.
No sources or sinks means no generation or consumption of mass inside the reservoir.
9. The density of the fluid is independent of position and time, since it is an incompressible liquid.
10. Kinetic energy can be ignored, and laminar flow ($Re < 10$) is assumed. [2]

3 Applicability of Darcy's equation

Darcy's Law can be applied to various scenarios such as:

- To study the seepage in earth drums by calculating the discharge rate through a porous medium.
- To describe oil, water and gas flows in petroleum reservoirs.
- To describe flow in porous media by taking into spatial deviations of pressure and velocity.
- To model groundwater such as the flow of water through an aquifer.

Moreover, Darcy's Law can only be applied under the condition of laminar flow. Hence the Reynolds number (Re) of the fluid in consideration must have a value between 1 and 10. [2]

4 Deriving the Continuity equation

Continuity equation for a single-phase flow, with constant density, flowing through a porous medium is given by:

$$\varepsilon \frac{\partial \rho}{\partial t} = -(\nabla \cdot \rho v), \quad (2)$$

where ε is the ratio of the pore volume to the total volume, also known as porosity, v is the superficial velocity or the volumetric flux, defined as “the volume rate of flow through a unit cross-sectional area of the solid plus fluid” [3], and $\nabla = \left(\frac{\delta}{\delta x}, \frac{\delta}{\delta y} \right)$ is the Del operator.

Darcy's Equation of motion is defined as

$$v = \frac{k}{\mu} (\nabla p - \rho g), \quad (3)$$

and Equations (2) and (3) can then be combined into

$$\left(\frac{\varepsilon \mu}{k} \right) \frac{\partial p}{\partial t} = (\nabla \cdot \rho (\nabla p - \rho g)), \quad (4)$$

where p is the pressure of the fluid, and g is the gravitational acceleration. Since density is independent of time and $\rho = \rho_0$ and ρg can be written as ϕ hence giving

$$0 = (\nabla \cdot (\nabla p - \nabla \phi)).$$

Since the gravitational effects on the fluid are neglected, the term $\nabla \phi$ is neglected:

$$\nabla^2 p = 0 \quad (5)$$

For our case of two-dimensional flow, Equation (5) simplifies to:

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = 0 \quad (6)$$

5 Shell Balance

By keeping the law of conservation of mass in mind, we apply mass balance on an infinitesimally small volume element to get

$$\varepsilon \rho v_x H \cdot \Delta y \Big|_x - \varepsilon \rho v_x H \cdot \Delta y \Big|_{x+\Delta x} + \varepsilon \rho v_y H \cdot \Delta x \Big|_y - \varepsilon \rho v_y H \cdot \Delta x \Big|_{y+\Delta y} = 0. \quad (7)$$

Dividing Equation (7) by $H \cdot \Delta x \cdot \Delta y$ and taking the limit to 0 gives

$$\lim_{\Delta x \rightarrow 0} \frac{\varepsilon \rho v_x \Big|_x - \varepsilon \rho v_x \Big|_{x+\Delta x}}{\Delta x} + \lim_{\Delta y \rightarrow 0} \frac{\varepsilon \rho v_y \Big|_y - \varepsilon \rho v_y \Big|_{y+\Delta y}}{\Delta y} = 0 \quad (8)$$

Since ε and ρ are constant, Equation (7) becomes:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0, \quad (9)$$

where v_x and v_y are the volumetric flux in the x - and y - directions respectively, given by:

$$v_x = -\frac{k}{\mu} \frac{\partial p}{\partial x} \quad (10)$$

$$v_y = -\frac{k}{\mu} \frac{\partial p}{\partial y} \quad (11)$$

Substituting equation (10) and (11) into (9) results in (6) as derived before:

$$-\frac{k}{\mu} \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} \right) = 0$$

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = 0$$

To obtain volumetric flux for the reservoir, the following equation is used

$$v = \sqrt{v_x^2 + v_y^2}. \quad (12)$$

To obtain the interstitial velocity u_x and u_y , the volumetric flux is divided by the porosity and the resultant velocity of the fluid can be written as:

$$u = \sqrt{\left(\frac{v_x}{\varepsilon}\right)^2 + \left(\frac{v_y}{\varepsilon}\right)^2} \quad (13)$$

Table 1 below summarizes all the major equations that will be used in the calculation of the various quantities within the reservoir.

Table 1: Summary of Equations used

Pressure Equation	$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = 0$
Volumetric flux equation	$v = \sqrt{v_x^2 + v_y^2}$
Velocity equation	$u = \sqrt{\left(\frac{v_x}{\varepsilon}\right)^2 + \left(\frac{v_y}{\varepsilon}\right)^2}$

6 Modeling using the Finite Difference Method

As discussed earlier, numerical simulation is used to calculate the pressures, volumetric flux and velocity of the fluid within the reservoir. The finite difference method was implemented into our methodology and a MATLAB code was written to obtain the results.

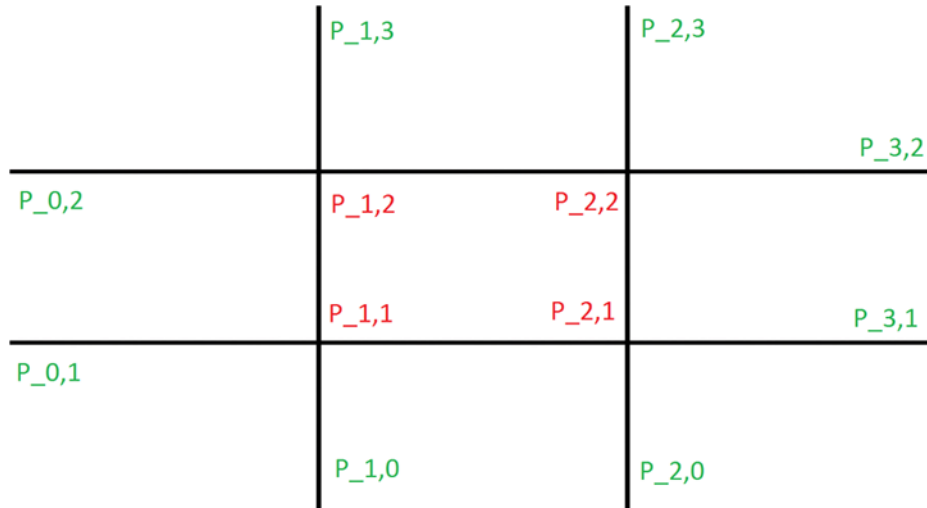


Figure 1: A small grid used to demonstrate the finite difference method

Based on nodal network, the finite difference derivation results in the following equations for the x - and y - directions:

$$\frac{\partial^2 p}{\partial x^2} = \frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2} \quad (14)$$

$$\frac{\partial^2 p}{\partial y^2} = \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2} \quad (15)$$

Substituting equations (14) and (15) in (6) results in the following equation:

$$p_{i+1,j} + p_{i-1,j} + p_{i,j+1} + p_{i,j-1} - 4p_{i,j} = 0. \quad (16)$$

Since the west side of our reservoir is impermeable the equation to predict the pressure at the impermeable side changes and results in:

$$2p_{i+1,j} + p_{i,j+1} + p_{i,j-1} - 4p_{i,j} = 0 \quad (17)$$

7 MATLAB Code

7.1 Boundary Pressure Fit

To allow for an increase in the resolution (i.e. more nodes per dimension) of a pressure grid, it is imperative to have known values of pressure along the boundaries. Since the given number of pressure values is fixed, one way to increase the resolution would be to first find the best fit polynomials for each boundary. Figure 2 shows the polynomial fits for each boundary, where the blue line is the continuous plot of the polynomial, and the red circles are the given fixed values in the prompt.

When increasing the number of variables between the boundaries, we can then find the intermediate boundary values using these polynomials, and then apply the finite difference method to calculate pressure, volumetric flux and fluid velocity.

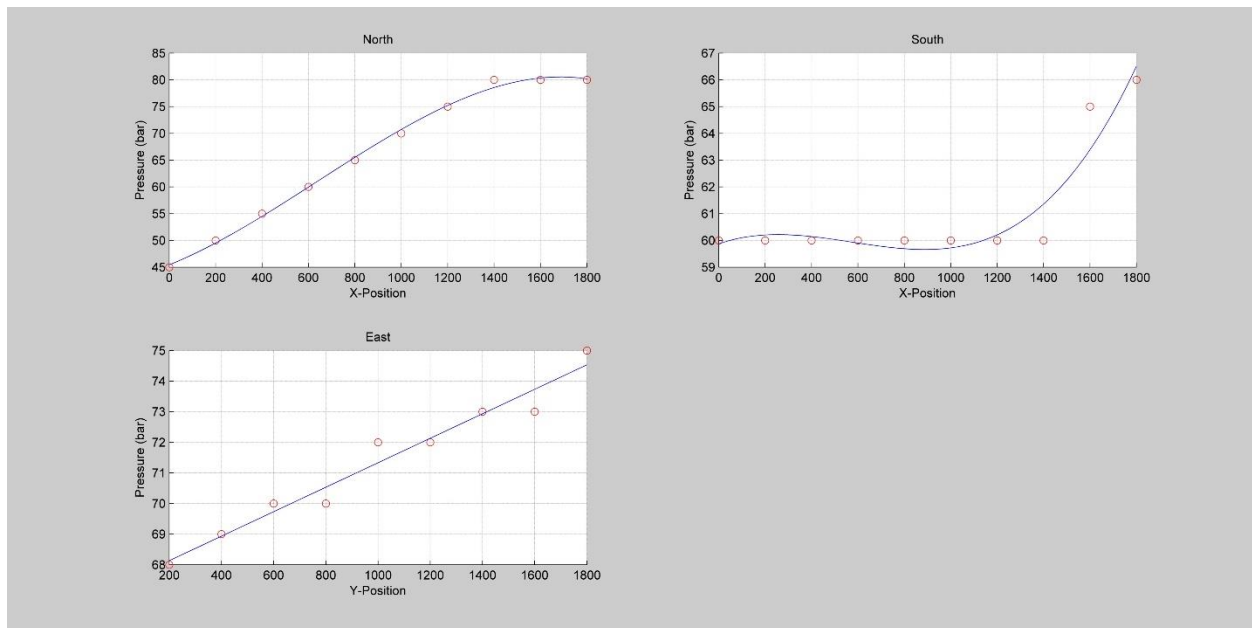


Figure 2: Best-fit polynomials for the North, South and East Boundaries. The West boundary is impermeable.

7.2 Results

7.2.1 10 nodes per dimension

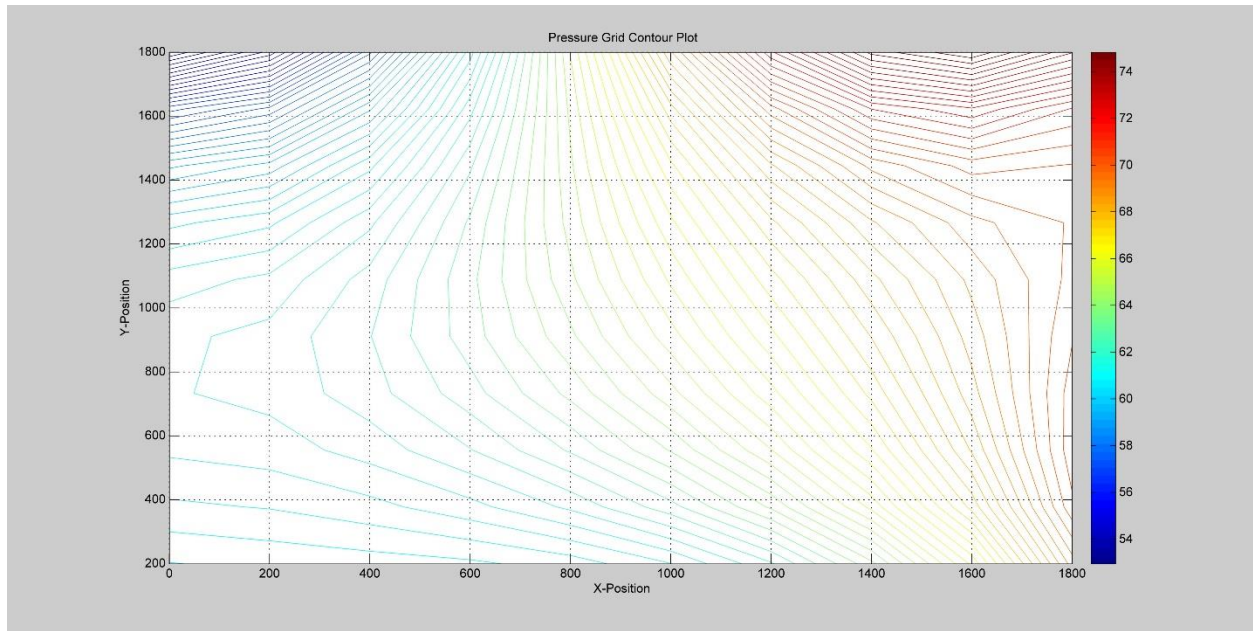


Figure 3: Pressure Contour Plot for 10 nodes per dimension

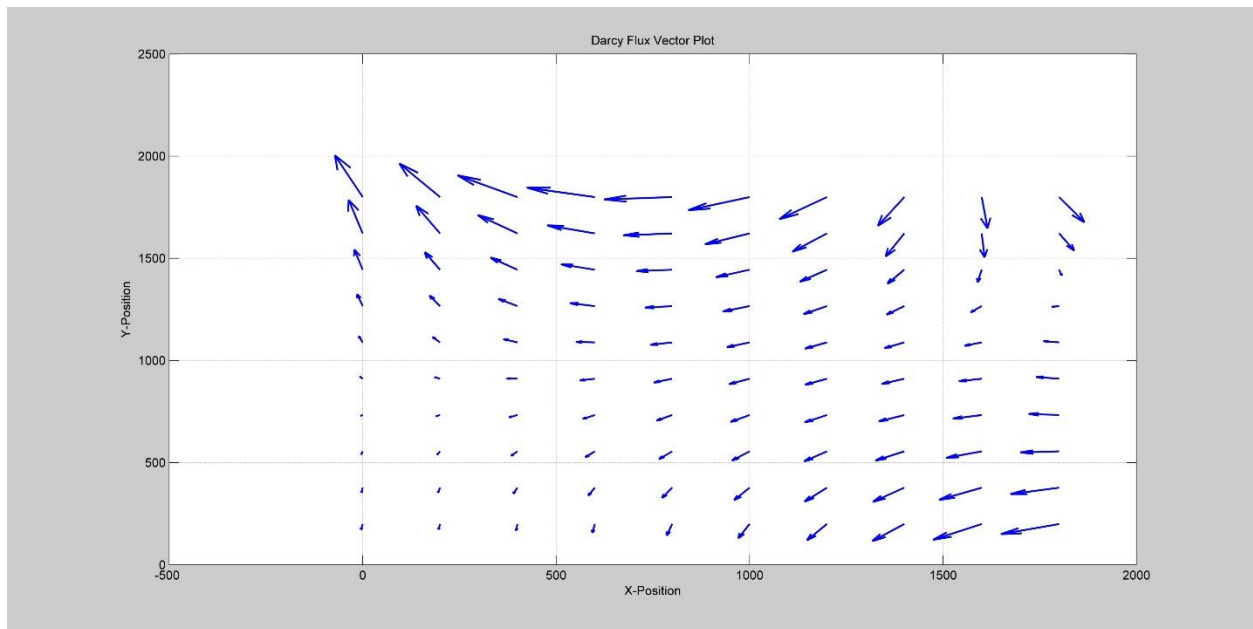


Figure 4: Darcy Flux vector plot for 10 nodes per dimension

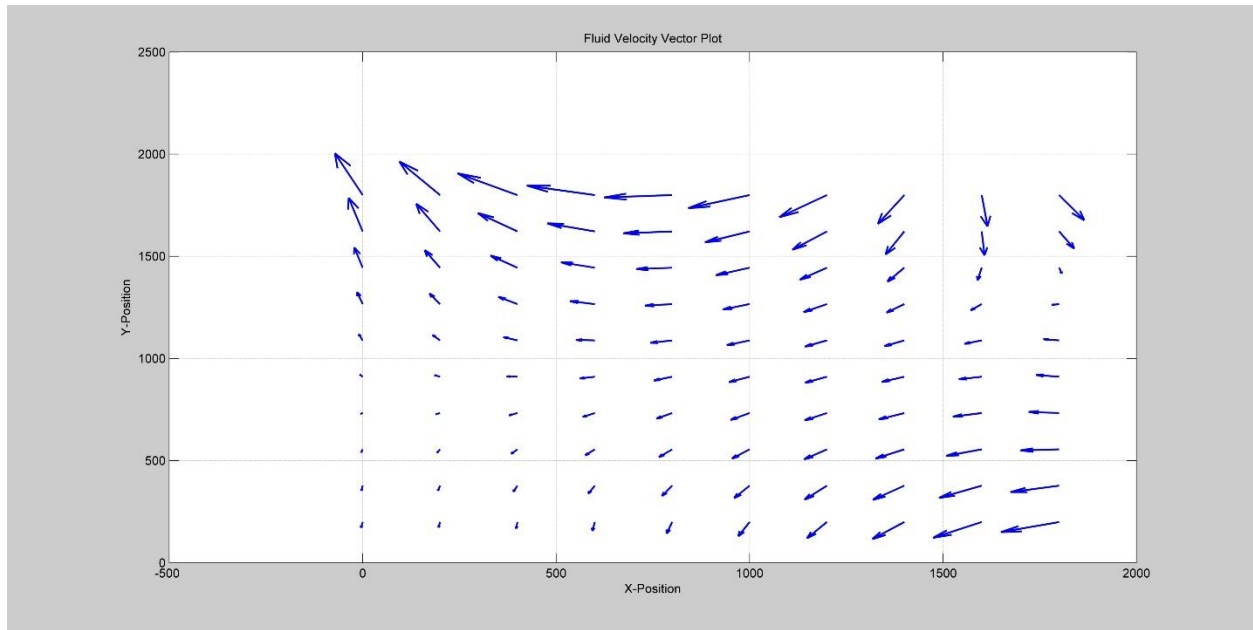


Figure 5: Fluid Velocity vector plot for 10 nodes per dimension

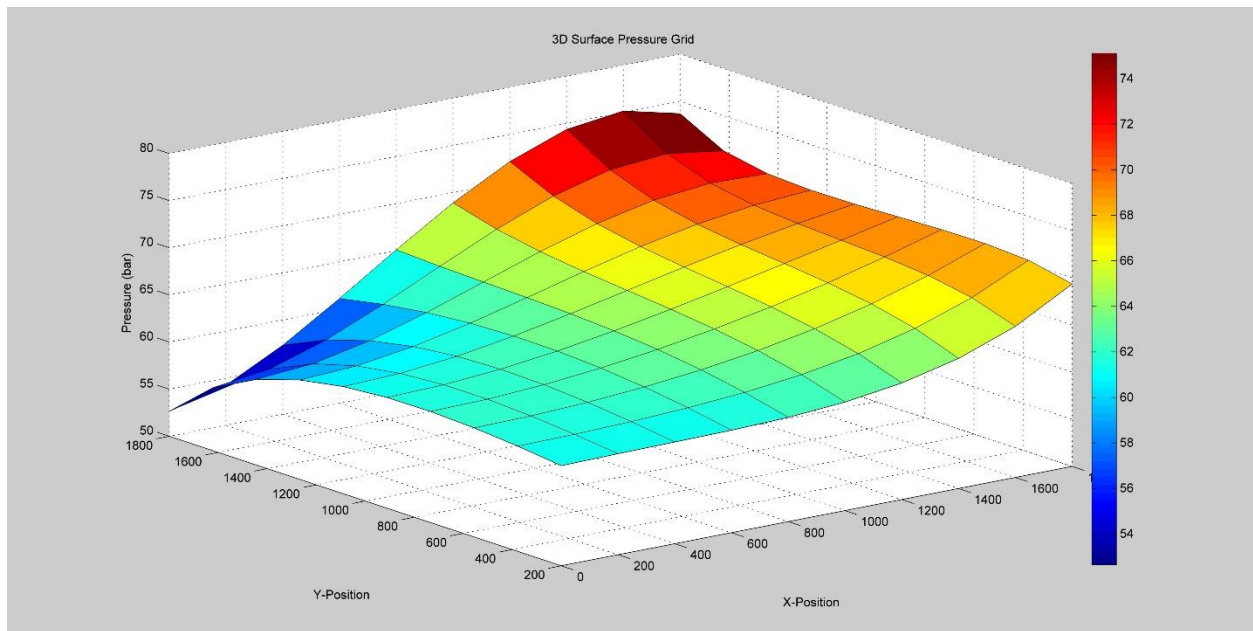


Figure 6: 3D Pressure Grid for 10 nodes per dimension

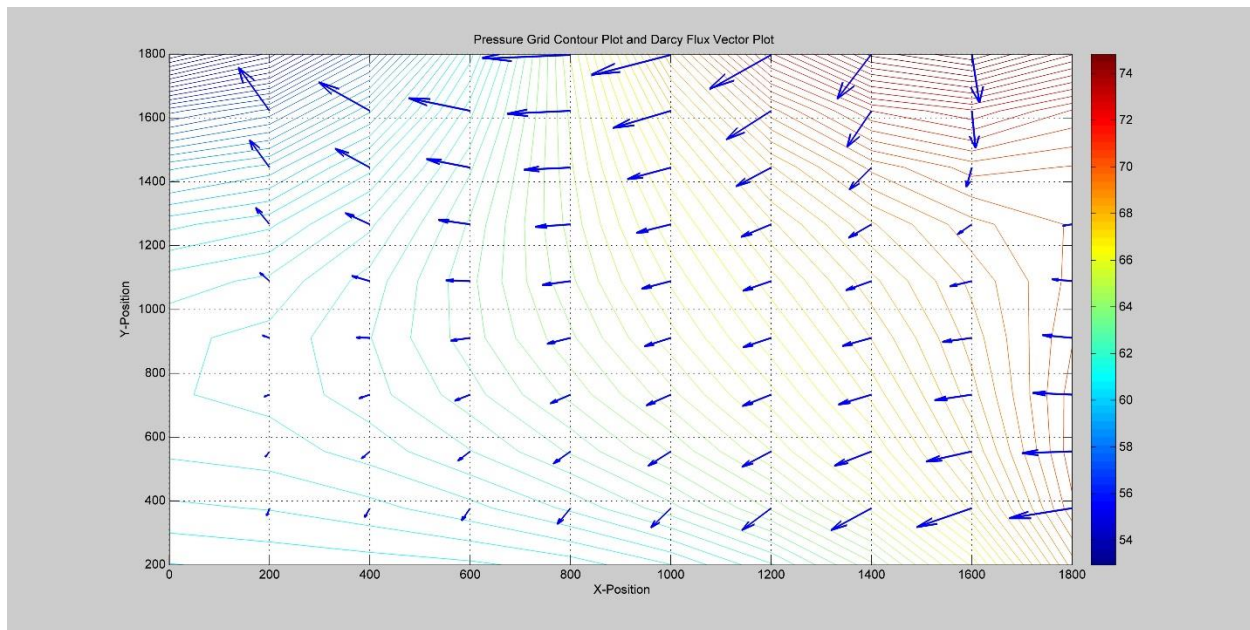


Figure 7: Superimposing the Darcy Flux vector plot on the Pressure contour plot for 10 nodes per dimension

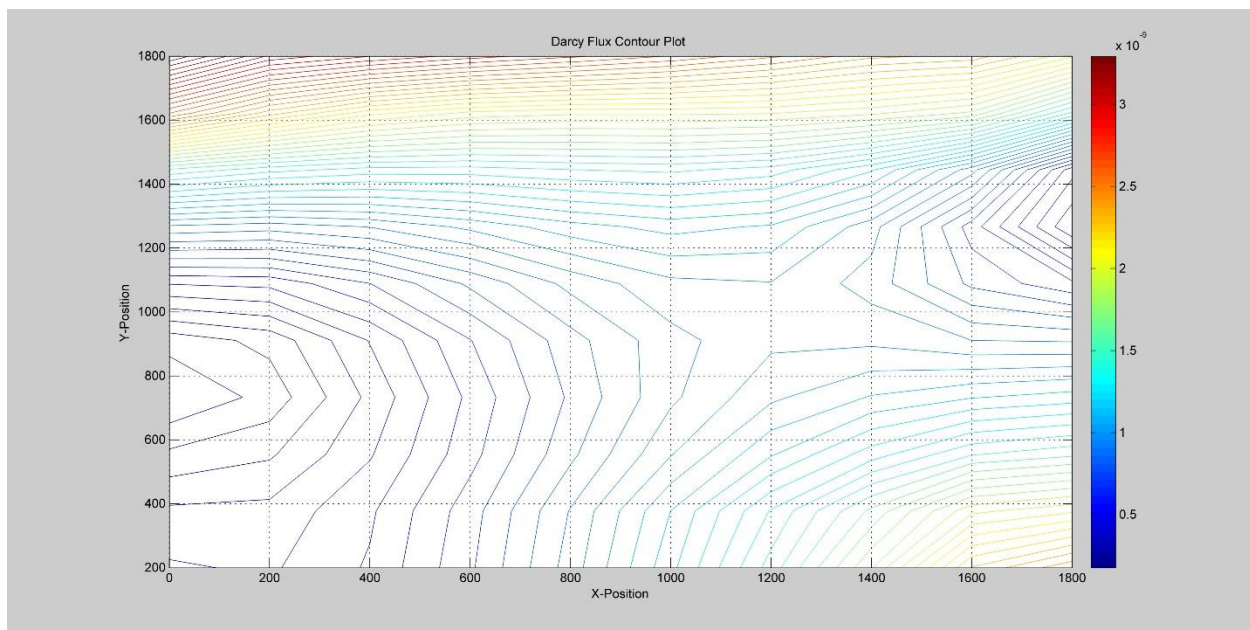


Figure 8: Darcy Flux contour plot for 10 nodes per dimension

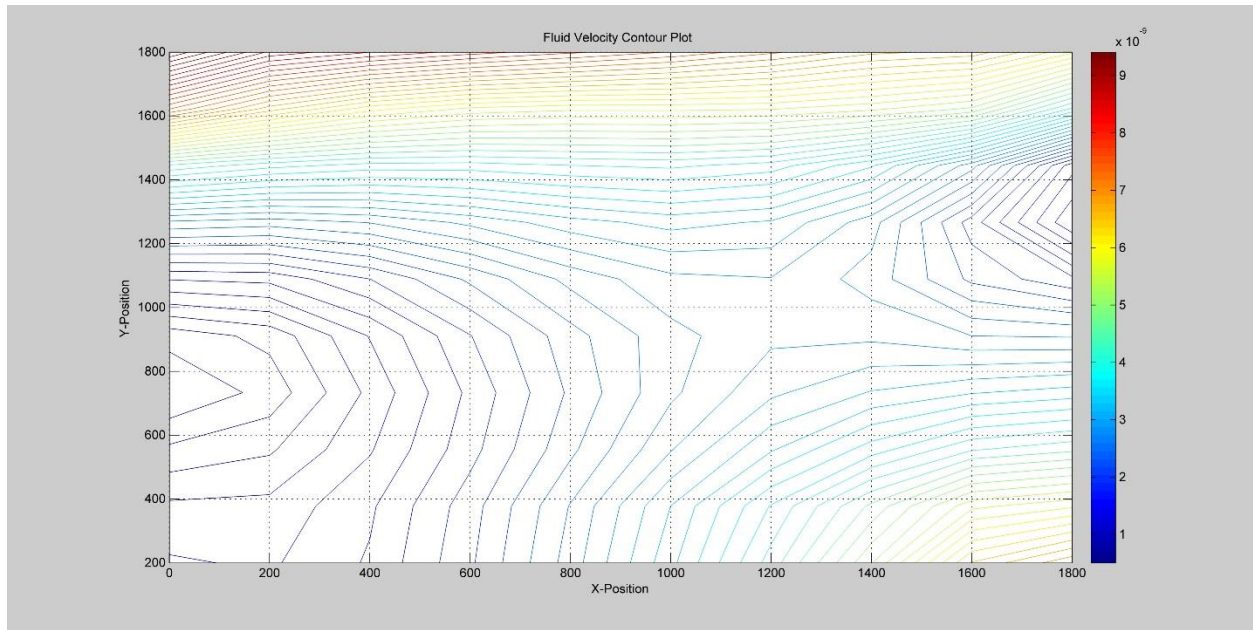


Figure 9: Fluid Velocity contour plot for 10 nodes per dimension

7.2.2 25 nodes per dimension

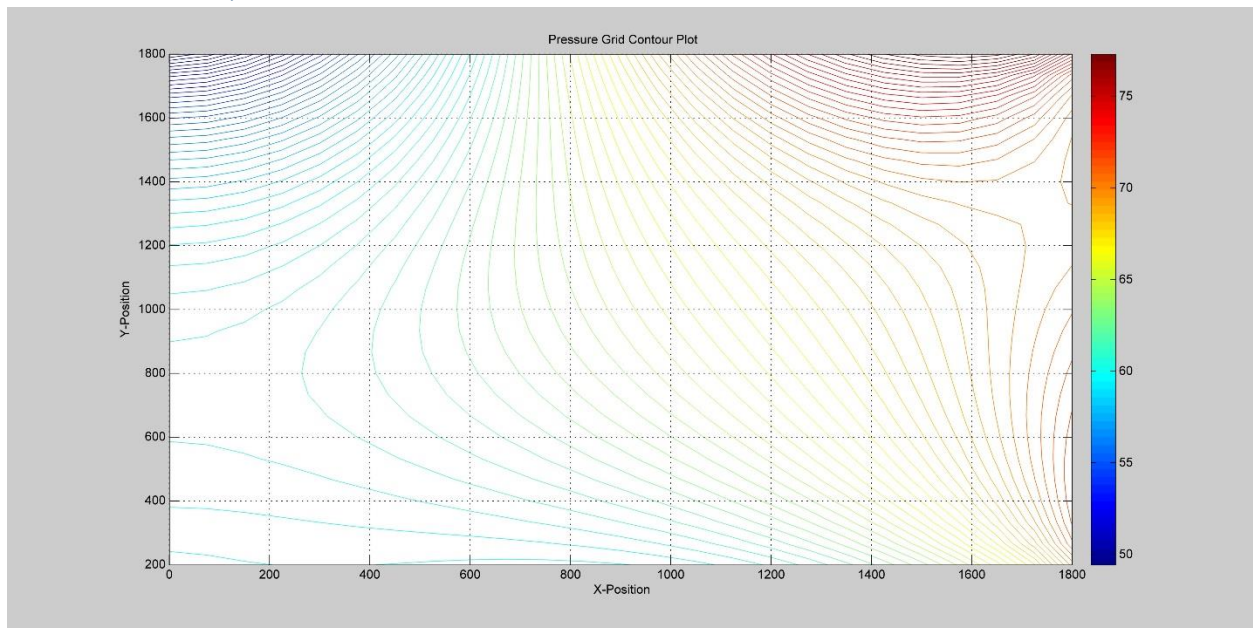


Figure 10: Pressure contour plot for 25 nodes per dimension

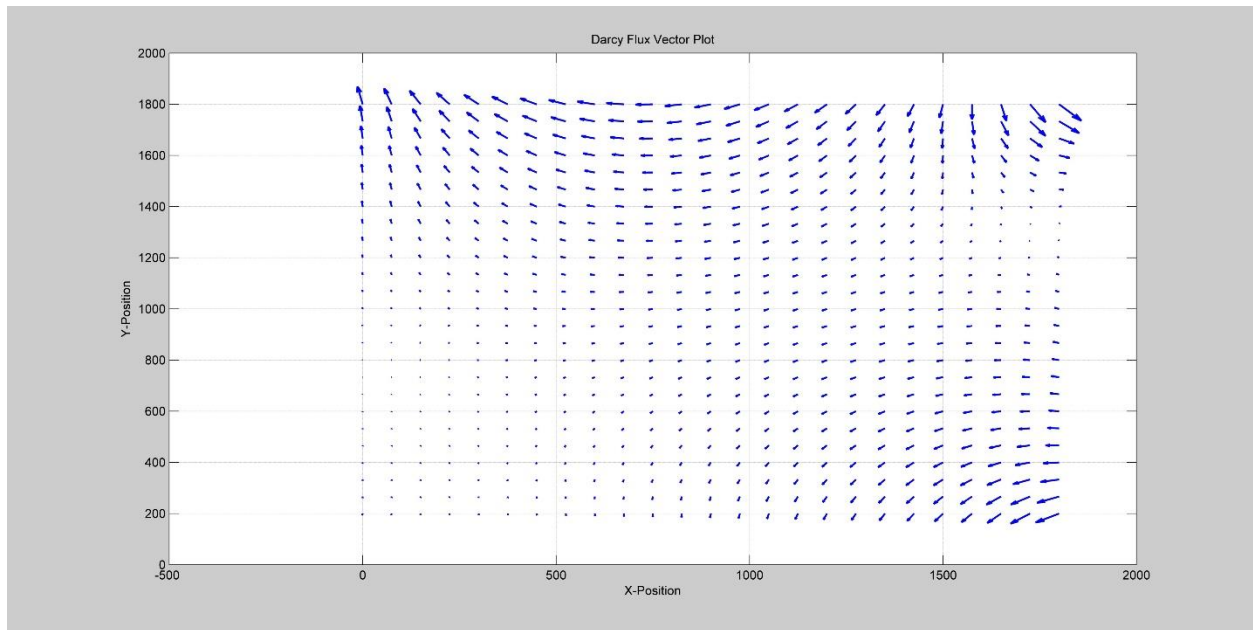


Figure 11: Darcy flux vector plot for 25 nodes per dimension

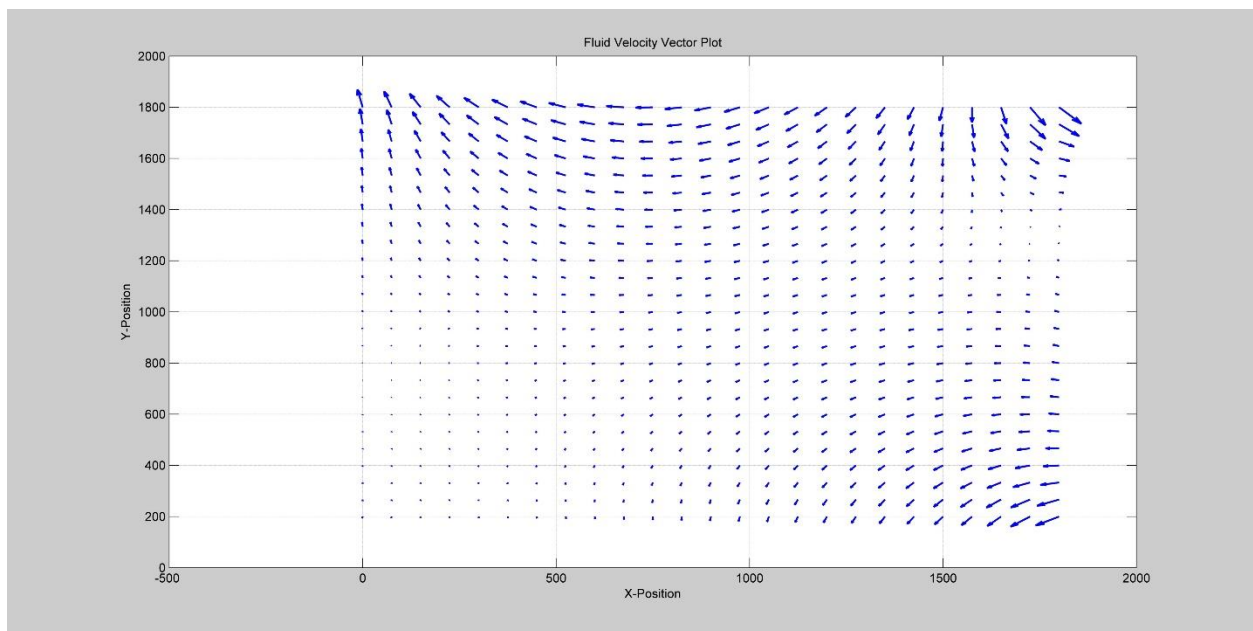


Figure 12: Fluid velocity vector plot for 25 nodes per dimension

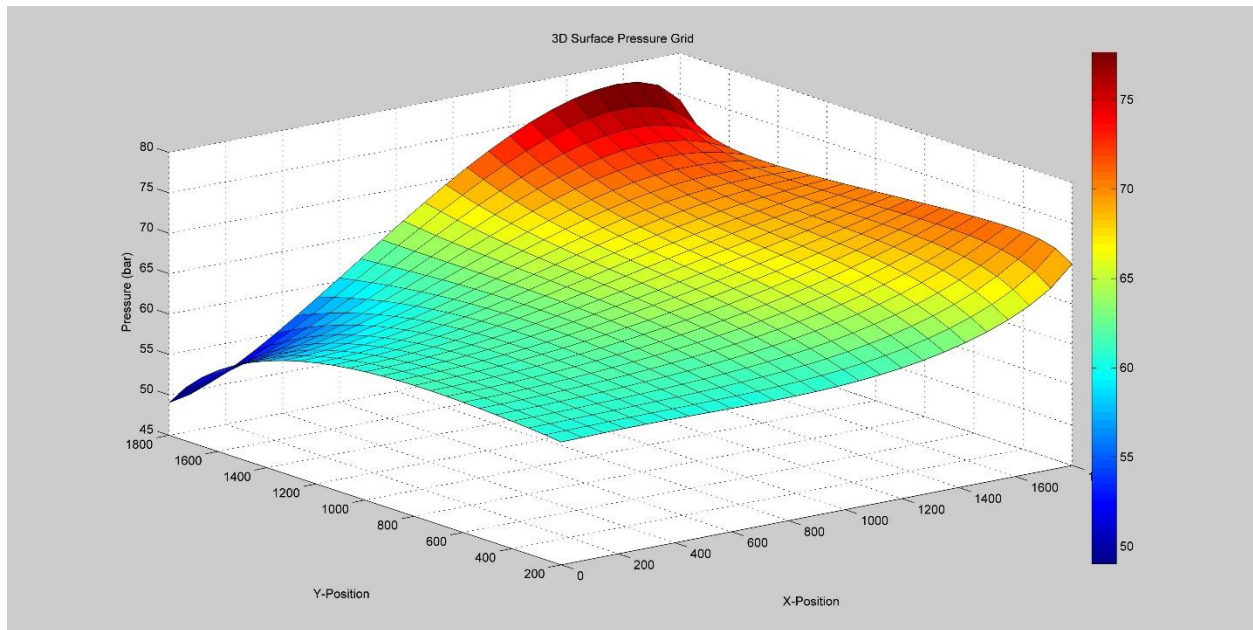


Figure 13: 3D Pressure Grid for 25 nodes per dimension

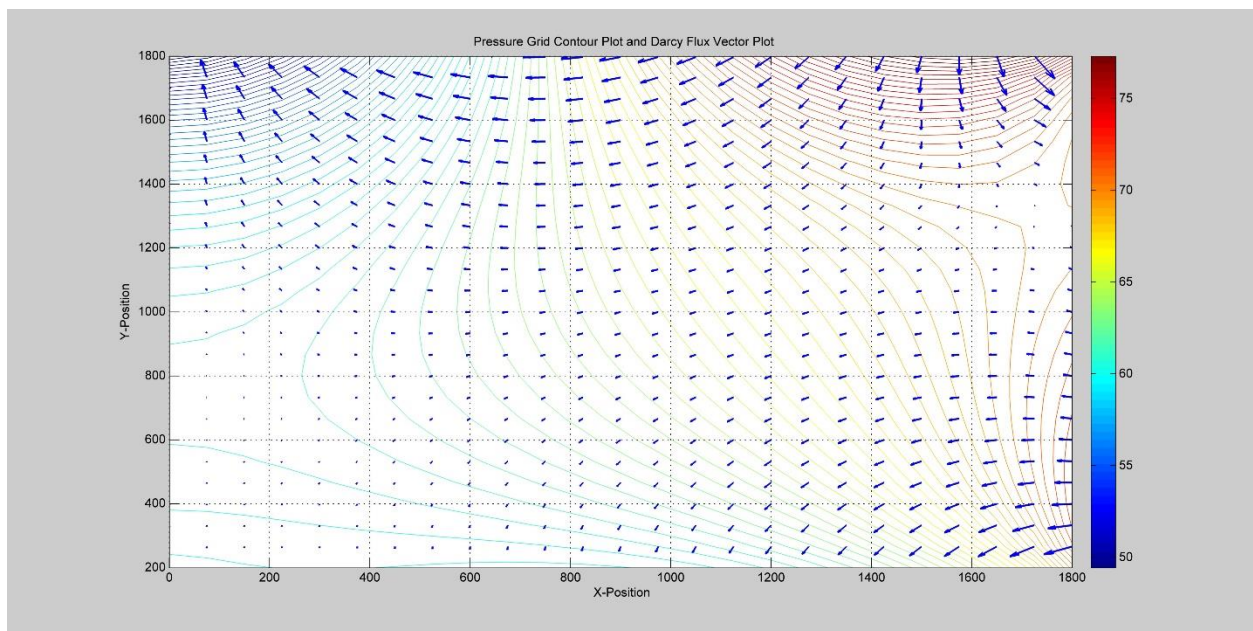


Figure 14: Superimposing the Darcy Flux vector plot on the Pressure contour plot for 25 nodes per dimension

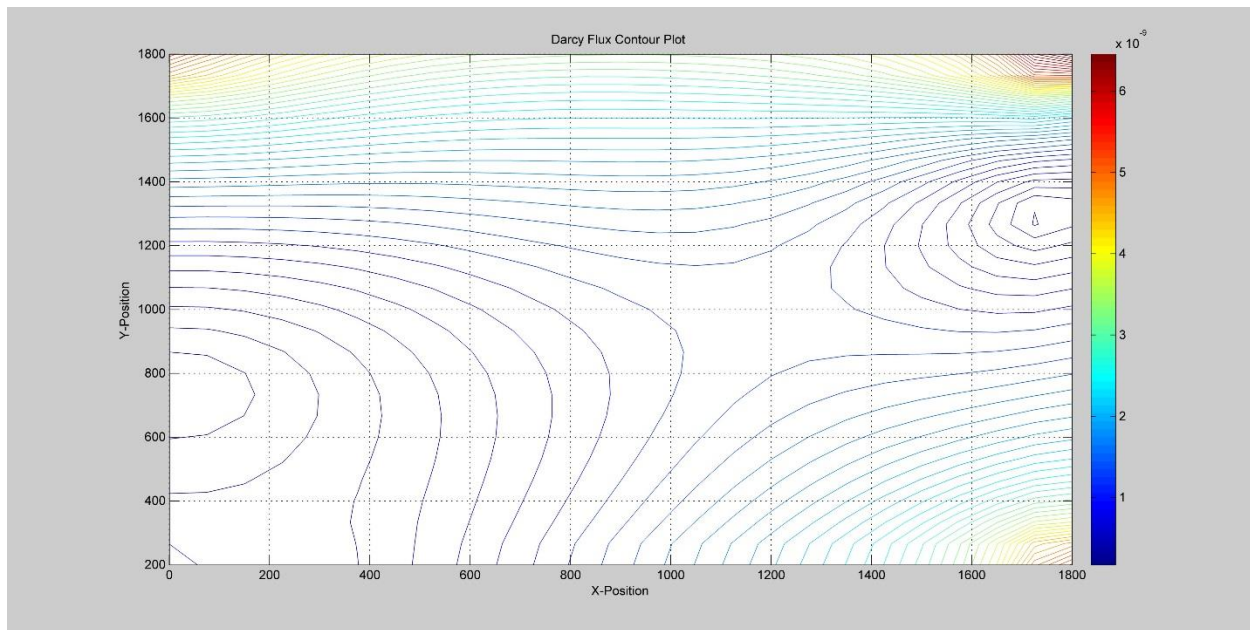


Figure 15: Darcy Flux contour plot for 25 nodes per dimension

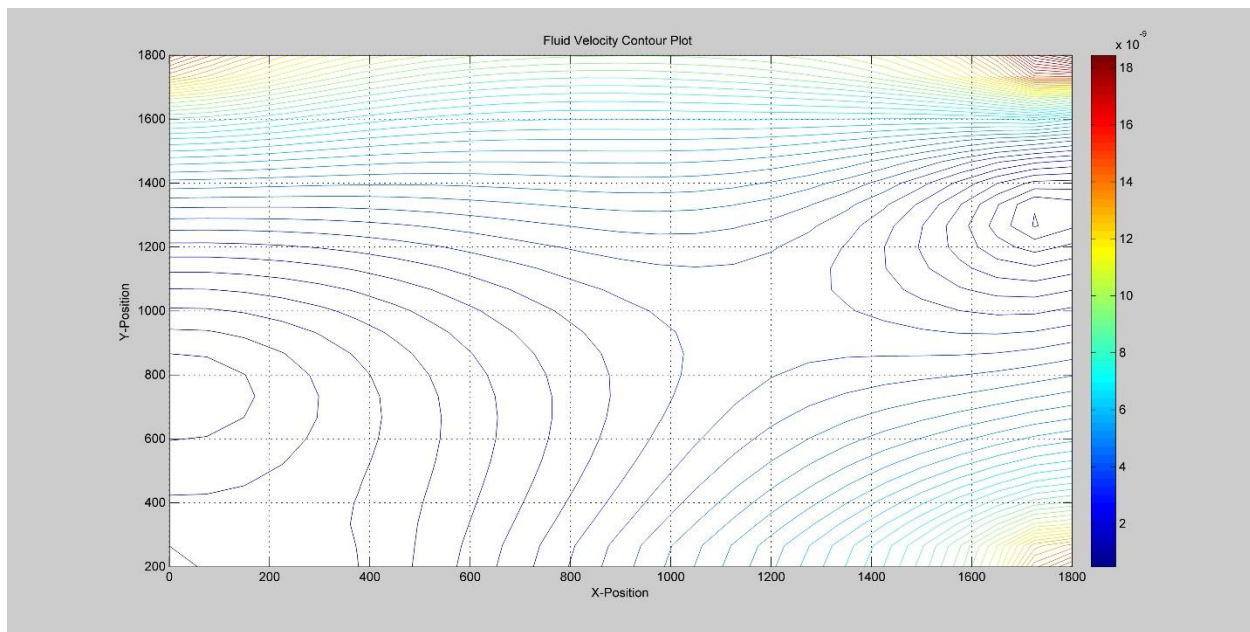


Figure 16: Fluid Velocity contour plot for 25 nodes per dimension

7.2.3 MATLAB Code Script

The MATLAB code script is attached to the end of the report, and is commented in detail for the benefit of the reader. In short, the MATLAB code runs through every node in the pressure grid, identifying the known and unknown pressure values immediately surrounding the current node. It then stores both values into respective matrices and calculates the pressure at each node in the grid. This method is not iterative, and hence, there was no error matrix to find. The only error that may arise would be from rounding errors in MATLAB itself, and these are typically in the order of $\sim 10^{-15}$. In other words, they are negligible enough to be ignored.

8 Analysis & Conclusion

To increase the resolution (i.e. the number of nodes per dimension) of the pressure grid, a polynomial fit of the pressure values given for the North, East and South boundaries was calculated using MATLAB and the results are shown in Figure 2. Even though the fitted polynomial (in blue) does not pass through all the given values (in red), it is good practice to not overfit data to avoid misleading results. The West boundary is impermeable, and the pressure values along the border are assumed to be unknown in the calculations and, thus, a part of the pressure grid to be computed.

Now that we can increase the resolution of the pressure grid at will, we computed the results for two dimensions: the 10x10 and the 25x25 nodes per dimension respectively. First, for the 10x10-pressure grid, the aim was to get a rough estimate of the shape of the pressure, velocity and volumetric contour plots and the respective vector plots. Figure 3 shows the contour plot for the 10x10 pressure grid, and as expected, the pressure is highest (dark red color and closely matching contour lines) in the northeast side and lowest (dark blue and closely matching contour lines) in the northwest side. This is verified by the given pressure values in the problem statement. Figures 4 and 5 show the Darcy flux and fluid velocity vector plots, which are expected to look very similar as shown by the equations in Table 1. Moreover they follow the same pattern as expected, the velocity and flux of the fluid move from higher pressure to the

direction of the lower pressure this is more evident when one looks at figure 7. At the northeast side where the pressure is highest, the fluid moves from northwest side to the northeast side. In addition, at the southeast side where the pressure is relatively high, the fluid tends to move in the south direction where the pressure is lower for that region. Figure 6 shows a three-dimensional representation of the pressure grid since it is easier to visualize the direction of the pressure changes that way. Finally, Figure 7 shows the Darcy flux plot superimposed on the contour plot to verify that it is correct, since it is expected for the flux to move from the direction of higher pressure to lower pressure in a reservoir. Figures 10 to 14 repeat the same results shown in Figures 3 to 7, but for the 25x25 pressure grid size. This is done to smoothen the respective visual representations and increase the accuracy of the pressure grid values, as defined by the finite difference method.

In conclusion, our results show that the finite difference method, in combination with matrices and their unique properties, can be effectively used to simulate a simple reservoir and calculate the inner pressure values to a high degree of accuracy and at a low computational cost. Moreover, this work supported the validity of Darcy's Law under the conditions given in the problem statement, and its usage to find not only the pressure values, but the volumetric (Darcy) flux and fluid velocity.

9 References

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